

BALLYKELLY, United KingdomWMO: **039080**Lat: **55.057N**Lon: **7.007W**Elev: **5**StdP: **101.26**Time Zone: **0.00 (EUW)**Period: **97-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-1.8	-5.4	2.4	-2.2	-3.6	2.8	0.1	17.1	7.7	15.1	8.7	1.3	120	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	21.9	17.3	20.5	16.3	19.3	15.7	18.2	20.9	17.2	19.6	16.3	18.4	4.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.1	19.6	16.1	11.4	18.6	15.3	10.8	17.7	51.3	21.0	48.3	19.6	45.8	18.5	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	13.0	11.4	10.0		-5.7	25.2	2.7	1.8	-7.7	26.4	-9.3	27.5	-10.8	28.5	-12.8	29.7
(5)					-6.2	20.0	2.7	1.0	-8.1	20.7	-9.7	21.3	-11.2	21.9	-13.2	22.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	5.3	5.7	6.9	8.7	11.2	13.4	15.0	14.8	13.6	10.6	7.7	5.2
	DBStd	4.44	3.04	2.80	2.60	2.70	2.35	2.36	2.04	1.87	2.24	2.64	2.67	3.62
	HDD10.0	694	148	121	102	56	12	2	0	0	1	26	76	151
	HDD18.3	3099	405	352	356	289	221	149	107	111	143	241	319	408
	CDD10.0	641	1	2	4	17	49	103	154	149	109	43	7	1
	CDD18.3	6	0	0	0	0	0	1	3	1	1	0	0	0
	CDH23.3	17	0	0	0	0	0	2	13	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	4.5	5.2	5.2	4.9	4.3	4.3	3.9	3.8	3.7	4.2	4.7	5.1	4.7
Precipitation	PrecAvg	1149	122	97	83	73	69	80	85	93	83	116	118	133
	PrecMax	1409	177	216	170	114	135	164	141	191	182	233	221	269
	PrecMin	884	27	34	24	23	13	23	33	15	15	42	44	57
	PrecStd	143	41	45	38	25	32	34	25	38	35	49	39	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	13.6	14.9	19.0	21.8	23.1	25.1	22.6	22.1	17.5	14.9	13.2
		MCWB	11.1	12.2	11.3	12.6	15.7	17.2	19.3	18.2	17.8	14.5	12.9	11.6
	2%	DB	11.9	12.4	13.4	16.5	19.5	21.0	22.1	20.7	20.0	16.2	13.5	11.9
		MCWB	10.4	11.0	10.8	12.0	14.5	16.2	17.6	17.1	16.4	14.0	12.0	10.6
	5%	DB	10.8	11.1	12.1	14.7	17.3	19.3	20.2	19.4	18.5	15.4	12.5	11.0
		MCWB	9.5	9.7	10.0	11.3	13.3	15.5	16.4	16.3	15.6	13.4	11.2	9.7
	10%	DB	9.7	10.1	11.0	13.3	15.6	17.7	18.7	18.2	17.2	14.5	11.6	9.9
		MCWB	8.4	8.6	9.2	10.4	12.2	14.5	15.5	15.5	14.7	12.6	10.4	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.7	12.2	14.1	17.0	18.6	19.5	19.2	18.7	15.6	13.2	11.9
		MCDB	12.6	13.4	13.6	16.8	21.1	21.8	24.0	21.5	20.9	16.7	14.4	13.0
	2%	WB	10.6	11.1	11.3	12.7	15.0	16.9	18.2	17.8	17.0	14.5	12.2	10.8
		MCDB	11.7	12.1	12.8	15.3	18.5	19.8	21.3	19.7	19.0	15.8	13.3	11.8
	5%	WB	9.6	9.9	10.4	11.6	13.7	15.7	17.0	16.9	16.0	13.7	11.4	9.8
		MCDB	10.7	11.0	11.8	14.3	16.6	18.6	19.6	18.8	18.0	15.1	12.3	10.9
	10%	WB	8.5	8.8	9.4	10.6	12.7	14.9	16.0	16.0	15.1	12.9	10.4	8.7
		MCDB	9.6	9.9	10.8	13.0	15.0	17.4	18.0	17.6	16.9	14.3	11.4	9.7
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.3	6.3	7.0	7.2	6.7	6.5	6.5	6.3	5.7	4.7	5.0
		MCDBR	5.2	5.2	7.4	8.8	9.7	9.3	9.1	8.1	7.7	6.1	5.1	5.4
		MCWBR	4.8	4.8	5.3	5.6	5.9	5.6	5.3	5.3	5.0	4.8	4.8	5.2
	5% WB	MCDBR	5.2	5.1	6.3	8.1	8.7	8.5	8.3	6.9	6.8	5.8	4.9	5.4
		MCWBR	4.9	5.0	5.0	5.5	5.8	5.7	5.3	5.1	5.0	5.0	4.8	5.4
Clear-Sky Solar Irradiance	taub	0.344	0.353	0.374	0.385	0.391	0.380	0.383	0.367	0.363	0.349	0.334	0.332	
	taud	2.198	2.277	2.270	2.282	2.310	2.377	2.375	2.433	2.436	2.392	2.265	2.170	
	Ebn at Noon	553	711	788	836	852	866	855	848	800	713	566	471	
	Edh at Noon	63	81	102	116	119	113	111	99	86	72	58	52	
All-Sky Solar Radiation	RadAvg	0.54	1.22	2.28	3.76	4.81	4.90	4.52	3.67	2.56	1.44	0.66	0.39	
	RadStd	0.06	0.11	0.20	0.30	0.43	0.48	0.34	0.30	0.14	0.13	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page